Bleeding and bleeding disorders

Approach and lab workup · VWD, hemophilia, rare factor and platelet disorders, acquired inhibitors · products, thresholds, massive hemorrhage, reversal · hemostatic agents and special situations

Compiled from the Fellowship Coagulation & Bleeding Disorders, Hemophilia and Transfusion Medicine reviews · ASH/ISTH/NHF/WFH 2021 VWD, WFH 2020, AABB 2023/2025, AABB/ICTMG 2025 guidance · FDA labels · primary trial reports verified against PubMed, October 2026.

APPROACH TO THE BLEEDING PATIENT

- **History:** how did surgery, dental extraction, childbirth, menses and trauma go? Transfusion or iron for bleeding counts. Drugs (antiplatelets, anticoagulants, SSRIs, fish oil), liver/renal disease, family history
- **ISTH-BAT abnormal at ≥4 in adult men, ≥6 in adult women, ≥3 in children** *Elbatarny 2014*. A low score in a young patient with few challenges does not exclude
- **Pattern: mucocutaneous** (epistaxis, gums, menses, petechiae) → **platelets or VWF**. **Deep tissue** (joints, muscle, delayed post-op) → **factor deficiency**. Delayed bleeding + poor healing → FXIII; fibrinolytic sites (mouth, nose, urinary) → FXI
- **First-line labs:** CBC + **smear**, **PT/INR, aPTT, Clauss fibrinogen, TT**; **VWF panel even if the aPTT is normal**
- **Pre-analytic traps:** heparin from a line, underfilled tube, Hct >55% (excess citrate) falsely prolong clotting times — redraw by clean venipuncture
- **Stanford:** Special Coag lab 650-723-4813 · Transfusion Medicine pager 12027

PT AND APTT PATTERNS

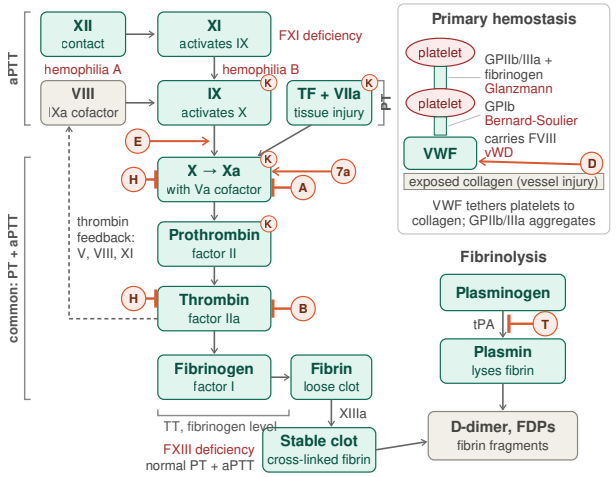
Pattern	Think
PT ↑ only	**FVII deficiency**; early vitamin K deficiency/warfarin (FVII t½ ~6 h); liver
aPTT ↑ only, bleeding	**FVIII, IX, XI deficiency**; VWD (low FVIII); **acquired hemophilia A**; heparin
aPTT ↑ only, no bleeding	**FXII**, prekallikrein, HMWK deficiency; **lupus anticoagulant** (thrombosis, not bleeding)
Both ↑	**FX, V, II or fibrinogen** deficiency; **DIC**; liver disease; severe vitamin K deficiency/warfarin; **DOACs**; dilution; amyloid (FX)
TT ↑	Heparin, **dabigatran**, low/dysfunctional fibrinogen, FDPs, paraprotein
Both normal + bleeding	**VWD**, platelet function defect or drug, **FXIII**, α2-antiplasmin/PAI-1, vascular (EDS, HHT, scurvy), mild factor deficiency

A normal PT does not exclude apixaban — use a drug-calibrated anti-Xa level.

SECOND-LINE TESTS

- **Specific factor assays** (one-stage, chromogenic for FVIII on emicizumab); **Bethesda** for inhibitors
- **VWF panel** ± multimers, low-dose RIPA (2B), FVIII-binding (2N)
- **Platelet function:** PFA-100 as a screen, **light transmission aggregometry**, flow for GPIIb/IIIa and GPIb, electron microscopy for granules
- **FXIII activity**, α2-antiplasmin, PAI-1 when screens are normal
- **Viscoelastic testing** (TEG/ROTEM) for acute bleeding; anti-Xa or drug levels for anticoagulants
- Normal full workup + convincing bleeding = **bleeding disorder of unknown cause**: manage as mild disorder (TXA ± DDAVP trial)

COAGULATION CASCADE



K vitamin K-dependent; warfarin blocks · H heparin + AT: inhibits IIa, Xa · A apixaban, rivaroxaban: Xa inhibitors · B dabigatran: direct IIa inhibitor · T TXA: blocks plasminogen lysine sites · D DDAVP: releases VWF + FVIII · 7a rFVIIa: activates X on platelets · E emicizumab: bridges IXa + X

Reversal: idarucizumab (dabigatran); 4F-PCC + vit K (warfarin); protamine (heparin)

PT = VII + common pathway; aPTT = XII, XI, IX, VIII + common; TT = fibrinogen → fibrin. Bleeding with normal PT + aPTT: vWD, platelet function defect, FXIII deficiency.

MIXING STUDY

- **1:1 with pooled normal plasma. Corrects → deficiency** (50% of any factor suffices). **Fails to correct → inhibitor**
- **Immediate inhibitor:** lupus anticoagulant, heparin (check TT / anti-Xa first)
- **Time- and temperature-dependent: anti-FVIII autoantibody**: corrects at first, prolongs after **1–2 h at 37°C** — ask the lab to incubate
- Next: factor levels, **Bethesda titer**, LA testing (dRVVT)

RARE FACTOR DEFICIENCIES

- **FXI:** autosomal, common in **Ashkenazi Jews**; **bleeding does not track the level**; fibrinolytic sites (mouth, nose, urinary). TXA ± plasma
- **FXIII: PT and aPTT normal.** Umbilical stump, delayed bleeding, poor healing, ICH. Quantitative FXIII activity (clot solubility is insensitive); monthly FXIII concentrate
- **FVII:** isolated ↑PT; bleeding correlates poorly with level; rFVIIa 15–30 μg/kg
- **Fibrinogen:** dysfibrinogenemia can bleed or clot (low Clauss, ↑TT); fibrinogen concentrate
- **α2-antiplasmin / PAI-1:** normal screens, delayed bleeding → TXA

VON WILLEBRAND DISEASE — ASH/ISTH/NHF/WFH 2021

- **Most common inherited bleeding disorder**; mucocutaneous bleeding, heavy menses, postpartum hemorrhage
- **Panel:** VWF:Ag, platelet-binding activity (**VWF:GPIbM** or GPIbR preferred over VWF:RCo), FVIII; **multimers** if activity:Ag <0.7
- **Diagnose type 1 at VWF <0.30 IU/mL regardless of bleeding, or <0.50 with abnormal bleeding.** Activity:Ag <0.7 → type 2
- **Levels move:** stress, inflammation, **pregnancy, estrogen raise VWF**; **group O runs ~25% lower**. Repeat before excluding; levels that normalize with age → reconsider, don't remove, the diagnosis
- **Rx: DDAVP trial** (VWF + FVIII at baseline, 1 h, 4 h) → responders use it for minor procedures · **TXA** for mucosal bleeding · **VWF concentrate** (plasma-derived or recombinant vonicog alfa) for type 3, 2B, non-responders, major surgery · hormonal therapy or TXA for heavy menses

VWD TYPES

Type	Defect · key pattern	Treatment
1 *~75%*	Partial quantitative; ratio ≥0.7, normal multimers	DDAVP usually works
2A	Loss of high-molecular-weight (HMW) multimers; ratio <0.7	Concentrate; DDAVP variable
2B	Gain-of-function GPIb binding → HMW loss + **low platelets**; ↑ low-dose RIPA; exclude platelet-type (GP1BA)	**Avoid DDAVP** (drops platelets); concentrate
2M	↓ GPIb binding with **normal multimers**; ratio <0.7	Concentrate; DDAVP variable
2N	**Low FVIII, near-normal VWF** — mimics mild hemophilia A but autosomal	FVIII-containing VWF concentrate
3	Near-absent VWF, FVIII <10%; joint bleeds	VWF concentrate

VWD — PROCEDURES AND PREGNANCY

- **Major surgery: VWF activity and FVIII both ≥0.50 IU/mL for ≥3 days**
- **Minor procedures:** raise VWF to ≥0.50 (DDAVP or concentrate) **+ TXA**; type 1 with VWF >0.30 and mild phenotype → **TXA alone**
- **Neuraxial anesthesia:** target VWF activity **0.50–1.50**; keep >0.50 while the catheter is in and ≥6 h after removal. Check levels in the third trimester
- **Postpartum: TXA** for type 1 or low VWF (reasonable in types 2 and 3)
- Give antiplatelets or anticoagulants when truly indicated; VWD is not a contraindication

ACQUIRED VON WILLEBRAND SYNDROME

- **Shear destroys HMW multimers: aortic stenosis** (Heyde: angiodysplasia bleeding, resolves after valve replacement), **LVAD** (near-universal), **ECMO**
- **Adsorption onto platelets: MPN with very high counts** (ET >1,000k) → hold aspirin, cytoreduce
- **Antibody / clearance: MGUS** and lymphoproliferative disease (IgG MGUS responds to IVIG), SLE, hypothyroidism
- Clue: low activity:Ag ratio, lost HMW multimers, short DDAVP response

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HEMOPHILIA A AND B

- **X-linked**; A (FVIII) ~1 in 5,000 males, B (FIX) ~1 in 25,000–30,000. Joints, muscle, ICH; ↑aPTT that corrects
- **Severity by factor level:** severe <1% (spontaneous bleeds) · moderate 1–5% · mild >5–<40% (bleeds with trauma, surgery). **Carriers can bleed** — check a level before surgery or delivery
- **Acute bleed: replace first, image second.** CNS/life-threatening **FVIII 80–100%, FIX 60–80%**, joint 40–60% *WFH 2020*. FVIII 1 U/kg → ↑~2%, FIX → ↑~1%
- **Inhibitors** (~25–30% of severe A, rare in B): ≥5 BU = high titer → **bypassing agents** (rFVIIa, aPCC); immune tolerance to eradicate
- **Emicizumab** (IXa–X bispecific): SC prophylaxis for A ± inhibitors (HAVEN 1, 3), **not for acute bleeds**. **Normalizes the aPTT and corrupts one-stage FVIII/Bethesda assays → use bovine chromogenic assays**
- Also: extended-half-life factor, anti-TFPI (**marstacimab, concizumab**), **fitusiran** (antithrombin siRNA), gene therapy. **DDAVP for mild A only**

ACQUIRED HEMOPHILIA A

- **Autoantibody to FVIII.** Elderly, **postpartum**, autoimmune, cancer; ~half idiopathic
- **New extensive soft-tissue bleeding** (ecchymoses, muscle, retroperitoneum); **joints rare**. ↑aPTT not correcting after incubation, low FVIII, Bethesda titer (does not predict bleeding)
- **Stop the bleed: recombinant porcine FVIII** (Obizur; 200 U/kg, then by FVIII level) or **bypassing agents** — rFVIIa 90 µg/kg q2–3 h or aPCC 50–100 U/kg (max 200 U/kg/day). **Emicizumab** prophylaxis is off-label in the US (approved in Japan 2022)
- **Eradicate at diagnosis:** prednisone 1 mg/kg ± cyclophosphamide or rituximab; seek the cause

PLATELET FUNCTION DISORDERS

Disorder	Defect · lab clue
Glanzmann	**GPIIb/IIIa** (ITGA2B/ITGB3), recessive; normal count. LTA: **no aggregation to any agonist except ristocetin**. Platelets (HLA-matched if alloimmunized); rFVIIa if refractory
Bernard-Soulier	**GPIb-IX-V**; **giant platelets + low count** (often mislabeled ITP). Absent ristocetin response, not corrected by normal plasma
Storage pool	δ-granule: absent secondary wave (Hermansky-Pudlak with albinism; Chediak-Higashi). α-granule: gray platelet syndrome (NBEAL2). DDAVP often helps
Acquired *most common*	Antiplatelets, uremia, MPN, bypass, SSRIs, paraprotein

PFA-100 catches VWD and severe defects, misses storage pool and mild defects; invalid if plt <100k or Hct <30%. **Light transmission aggregometry** is the reference test; bleeding time is obsolete.

BLOOD PRODUCTS

Product	Adult dose → effect
RBCs	~300 mL, Hct ~55–65%. **1 unit → Hb ↑ ~1 g/dL** in a non-bleeding adult. Single units, then reassess
Platelets	1 apheresis unit ≈ pool of 4–6 whole-blood units. **1 unit → ↑ ~30k** (less with fever, sepsis, splenomegaly, refractoriness). Dose doesn't change bleeding (PLADO)
Plasma	All coagulation factors. **10–15 mL/kg** (~4 units in 70 kg) for a real coagulopathy; TACO risk
Cryo	Fibrinogen, FVIII, VWF, FXIII. **10 units (2 pools of 5) → fibrinogen ↑ ~70–100 mg/dL**

RBC THRESHOLDS — AABB 2023

Setting	Transfuse at Hb
Hospitalized, hemodynamically stable (incl. ICU)	**<7 g/dL** (TRICC) *strong*
Heme/onc inpatients (chemo, HSCT)	**<7 g/dL** *conditional*
Cardiac surgery	**<7.5 g/dL**
Orthopedic surgery or CVD	**<8 g/dL** (FOCUS)
Septic shock	**<7 g/dL** (TRISS)
Acute MI + anemia	**Liberal <10 g/dL** *AABB 2025 conditional; ACC/AHA 2025 2b.* MINT: MI/death 16.9% restrictive vs 14.5% liberal, RR 1.15, P=0.07 — favored, not proven
Active bleeding	Clinical endpoints, not a number

PLATELET THRESHOLDS — AABB/ICTMG 2025

Setting	Transfuse below
Chemo or allo-HSCT, not bleeding	**10k** *strong; Rebulla, TOPPS*
Auto-HSCT or aplastic anemia, not bleeding	**No routine prophylaxis**
Consumptive, no major bleeding; CVC (compressible)	**10k**
Fever / sepsis	**~20k** *convention*
Lumbar puncture; low-risk IR	**20k**
High-risk IR; major non-neuraxial surgery	**50k**
Neurosurgery, ocular, neuraxial; CNS bleed	**~100k** *convention*
Active bleeding	**>50k**

Avoid platelets in TTP/HIT unless life-threatening bleeding; ITP only for serious bleeding, with IVIG + steroids; useless in PTP; **harmful in antiplatelet ICH** (PATCH).

MASSIVE HEMORRHAGE

- **Activate the MTP**; T&S, CBC, PT/aPTT, fibrinogen, iCa, ABG, lactate every 30–60 min
- **1:1:1 plasma : platelets : RBC** — PROPPR vs 1:1:2: 30-d mortality not different (22.4% vs 26.1%), but **hemostasis 86% vs 78%** and **exsanguination death at 24 h 9.2% vs 14.6%**
- **TXA 1 g over 10 min + 1 g over 8 h, within 3 h** (CRASH-2). **After 3 h bleeding deaths rose (RR 1.44) — don't start late**
- **Fibrinogen >150–200 mg/dL** (cryo or concentrate); replace **ionized calcium** early (citrate); prevent **hypothermia** and correct **acidosis**
- **Viscoelastic testing** (TEG/ROTEM): long R/CT → plasma; low MA/MCF → platelets or fibrinogen (FIBTEM); high LY30 → TXA
- Reverse anticoagulants; prehospital plasma in long air transport (PAMPer)

ANTICOAGULANT AND ANTIPLATELET REVERSAL

Agent	Reversal for major bleeding
Warfarin	**4F-PCC** 25 / 35 / 50 U/kg for INR 2–<4 / 4–6 / >6 (max 2500 / 3500 / 5000 U) or fixed 1000–1500 U, **+ vitamin K 10 mg IV**. PCC beats plasma: rapid INR correction 62% vs 10% (Sarode 2013); in ICH, INR ≤1.2 by 3 h 67% vs 9% (INCH). No bleeding, INR >10 → oral vitamin K
Dabigatran	**Idarucizumab 5 g IV** (REVERSE-AD: median reversal 100%); dialyzable. Normal TT excludes a relevant level
Apixaban, rivaroxaban, edoxaban	**4F-PCC 25–50 U/kg** or fixed 2000 U (off-label). **Andexanet left the US market 22 Dec 2025** — FDA judged thrombotic risk to outweigh benefit (ANNEXA-I: thrombosis 10.3% vs 5.6%, ischemic stroke 6.5% vs 1.5%). Charcoal if ingested <2–4 h
UFH	**Protamine 1 mg per 100 U** given in the last 2–3 h (max 50 mg); check aPTT
LMWH	Enoxaparin ≤8 h: **protamine 1 mg/mg**; 8–12 h: 0.5 mg/mg (max 50 mg). **Partial** (~60% of anti-Xa)
Antiplatelets	**No platelet transfusion in ICH** (PATCH: death or dependence OR 2.05) **unless neurosurgery**; DDAVP 0.4 µg/kg ×1 an option (NCS/SCCM 2016)
Thrombolytics	Stop; **cryo 10 units** (fibrinogen ≥150); **TXA 1 g** or aminocaproic acid 4–5 g

PLASMA AND CRYO — WHEN

- **Plasma for:** bleeding + coagulopathy (liver, DIC, dilution), massive transfusion, TTP plasma exchange, a factor deficiency without a concentrate (FV)
- **Don't correct an INR <1.7 with plasma before a procedure:** little effect, added TACO and TRALI risk. Not a volume expander; **not for warfarin reversal when PCC is available**
- **Cryo or fibrinogen concentrate: fibrinogen <150 mg/dL with bleeding** (DIC, MTP, APL; <200 in obstetric hemorrhage)

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HEMOSTATIC AGENTS — DOSING

Agent	Dose	Use · cautions
Tranexamic acid	**1 g IV over 10 min** (trauma + 1 g over 8 h; PPH repeat at 30 min if bleeding). PO 1300 mg TID × 5 d (menses); 5% mouthwash. Renal dosing. Aminocaproic acid 4–5 g then 1 g/h is the alternative	Mucosal, dental, menses, trauma, PPH, surgery. **Avoid in upper urinary tract bleeding** (ureteral clots). **No benefit in GI bleeding, more VTE** (HALT-IT)
Desmopressin (DDAVP)	**0.3 µg/kg IV/SC over 30 min** (max 20 µg); peak 30–60 min	Type 1 VWD, mild hemophilia A, carriers, uremia, platelet dysfunction. **Tachyphylaxis after 2–3 doses**; **restrict fluids 24 h, check Na**; caution CAD, age <2. **Avoid in 2B**
Cryo / fibrinogen concentrate	Cryo 10 units → ↑70–100 mg/dL. Concentrate (RiaSTAP, Fibryga): mg/kg = (target – measured mg/dL) ÷ 1.7 (RiaSTAP); ÷ 1.8 Fibryga (÷ 1.4 if <12 y)	Concentrate: standardized, pathogen-reduced, low volume
4F-PCC (Kcentra, Balfaxar)	25–50 U/kg by INR (reversal table)	II, VII, IX, X + C/S + heparin — **avoid in HIT**
aPCC (FEIBA)	50–100 U/kg q6–12 h, **max 200 U/kg/day**	Inhibitors (congenital or acquired). **On emicizumab: >100 U/kg/24 h for ≥24 h → TMA, thrombosis**
rFVIIa (NovoSeven RT)	Inhibitors / AHA: **90 µg/kg q2–3 h**. FVII deficiency: 15–30 µg/kg q4–6 h	Licensed: hemophilia with inhibitors, FVII deficiency, Glanzmann refractory to platelets, acquired hemophilia. **Off-label rescue: no mortality benefit, ↑ arterial thrombosis**
Vitamin K	**10 mg IV** slowly for bleeding; 1–5 mg PO non-urgent	IV acts in hours (full 12–24 h); avoid SC. Rare anaphylactoid reaction
FVIII / FIX	FVIII U = kg × desired % rise ÷ 2. FIX U = kg × desired % rise (÷ ~0.8 for some recombinant FIX)	Recheck trough; extended-half-life products dosed less often
Emicizumab	3 mg/kg SC weekly ×4, then **1.5 mg/kg weekly, 3 mg/kg q2wk or 6 mg/kg q4wk**	Prophylaxis only; breakthrough → rFVIIa preferred
Conjugated estrogens	**0.6 mg/kg/day IV × 5 d**	Uremic bleeding; onset ~1 d, effect lasts ~2 wk

HEAVY MENSTRUAL BLEEDING

- **Screen for a bleeding disorder** when heavy menses date from menarche, or with iron deficiency or postpartum hemorrhage
- **VWD:** hormonal therapy (combined contraception or levonorgestrel IUS) **or TXA**, preferred over DDAVP *ASH/ISTH 2021*
- Treat iron deficiency in parallel (IV iron if oral fails)

PATIENTS WHO DECLINE BLOOD

- **Ask product by product** — many Jehovah's Witnesses accept albumin, factor concentrates, PCC, fibrinogen concentrate or cell salvage; document it
- **Build reserve, cut losses:** IV iron ± ESA, TXA, pediatric tubes and fewer draws, early reversal of anticoagulants

SPECIAL SITUATIONS

Setting	Approach
Liver disease	**Rebalanced hemostasis**; INR does not predict procedural bleeding — **don't give plasma to fix it.** Fibrinogen and platelets matter more when bleeding; TEG/ROTEM. TPO-RA (avatrombopag, lusutrombopag) before procedures if plt <50k. Variceal bleeding is portal pressure — avoid volume
Uremia	Dialysis; raise Hb toward ~10 (improves platelet margination); DDAVP 0.3 µg/kg for acute bleeds or procedures; conjugated estrogens for longer effect; cryo if refractory
DIC	**Treat the cause.** Products only for bleeding or procedures: **plt >50k, fibrinogen >150 mg/dL**, plasma 15–30 mL/kg for prolonged PT/aPTT. ISTH score ≥5 = overt. Heparin only when thrombosis dominates
Cardiac surgery / ECMO	TXA routinely; protamine for heparin rebound; viscoelastic-guided algorithms reduce transfusion; bypass causes platelet dysfunction; **acquired VWS** on ECMO/LVAD
Obstetric hemorrhage	**TXA 1 g IV within 3 h** (WOMAN: bleeding death 1.5% vs 1.9%, RR 0.81; ≤3 h RR 0.69). **Fibrinogen ≤2 g/L predicts severe PPH** (PPV 100%, Charbit 2007) → keep >2 g/L. Abruption and amniotic fluid embolism → DIC
APL	Coagulopathy is the early killer: **start ATRA on suspicion**; keep fibrinogen >100–150 mg/dL and plt >30–50k with cryo and platelets; avoid invasive procedures
ICH	Reverse now: **4F-PCC + vitamin K** for warfarin; idarucizumab or PCC for DOACs. SBP target ~140, avoid <130 *AHA/ASA 2022*. No platelets for antiplatelet ICH (PATCH). **rFVIIa not recommended** (FAST, FASTEST)
Trauma	TXA ≤3 h (CRASH-2); TBI: head-injury death ↓ in mild–moderate injury (CRASH-3, RR 0.78); 1:1:1 (PROPPR); prehospital plasma (PAMPer)
Surgery in hemophilia	Major surgery: **FVIII 80–100% / FIX 60–80% pre-op**, then taper over 1–2 wk (WFH 2020); dental: TXA mouthwash + factor. Plan with the hemophilia treatment center

ISTH OVERT DIC SCORE

Parameter	0	1	2	3
Platelets (×10^9/L)	>100	50–100	<50	
D-dimer / FDP	normal		moderate ↑	strong ↑
PT prolongation	<3 s	3–6 s	>6 s	
Fibrinogen	>1 g/L	<1 g/L		

Score ≥5 = overt DIC; repeat daily. Only in a patient with a condition known to cause DIC *Taylor 2001*.

KEY TRIALS

Trial	Result → practice
CRASH-2 *Lancet 2010*	Trauma TXA (n=20,211): death 14.5% vs 16.0%, RR 0.91; harm if started >3 h
CRASH-3 *Lancet 2019*	TBI TXA ≤3 h: head-injury death 18.5% vs 19.8% (NS); mild–moderate RR 0.78
PROPPR *JAMA 2015*	1:1:1 vs 1:1:2: mortality NS; hemostasis 86% vs 78%, exsanguination 9.2% vs 14.6%
PAMPer *NEJM 2018*	Prehospital plasma, air transport: 30-d mortality 23.2% vs 33.0%
WOMAN *Lancet 2017*	PPH TXA (n=20,060): bleeding death 1.5% vs 1.9% (RR 0.81); no ↑ thrombosis
HALT-IT *Lancet 2020*	GI bleeding TXA: bleeding death 4% vs 4% (RR 0.99); VTE 0.8% vs 0.4% → **don't use**
PATCH *Lancet 2016*	Platelets for antiplatelet ICH: death/dependence OR 2.05 → **don't transfuse**
FAST / FASTEST *NEJM 2008 / Lancet 2026*	rFVIIa in ICH: less hematoma growth, no functional benefit; arterial events 9% vs 4%; FASTEST stopped for futility, thromboembolism RR 3.41
Sarode / INCH *2013 / 2016*	4F-PCC vs plasma for warfarin bleeding: hemostasis 72% vs 65% (non-inferior), rapid INR correction 62% vs 10%; ICH: INR ≤1.2 in 3 h 67% vs 9%
TRICC *NEJM 1999*	ICU Hb 7 vs 10: 30-d mortality 18.7% vs 23.3% (P=0.11) → restrictive is safe
FOCUS *NEJM 2011*	Hip fracture + CV risk, Hb 10 vs <8/symptoms: death or inability to walk 35.2% vs 34.7%
TRISS *NEJM 2014*	Septic shock, Hb 7 vs 9: 90-d mortality 43.0% vs 45.0%
MINT *NEJM 2023*	Acute MI, restrictive vs liberal <10: MI/death 16.9% vs 14.5%, RR 1.15, P=0.07
Rebulla *NEJM 1997*	AML induction, plt 10k vs 20k: major bleeding 21.5% vs 20%; 21.5% fewer transfusions
PLADO *NEJM 2010*	Platelet dose 1.1 / 2.2 / 4.4 ×10^{11}/m²: grade ≥2 bleeding 71 / 69 / 70%
TOPPS *NEJM 2013*	No prophylactic platelets: WHO ≥2 bleeding 50% vs 43% → keep 10k prophylaxis
REVERSE-AD *NEJM 2017*	Idarucizumab: median maximum dabigatran reversal 100%
ANNEXA-I *NEJM 2024*	Andexanet vs usual care, Xa-inhibitor ICH: hemostasis 67% vs 53%; thrombosis 10.3% vs 5.6% → withdrawn in US
HAVEN 1 / 3 *NEJM 2017 / 2018*	Emicizumab: annualized bleeds 2.9 vs 23.3 with inhibitors (–87%); 1.5 vs 38.2 without (–96%)